Engineering Mathematics - I · Unit_03 · Unit 3

DRAFT — UE26MA141A · Unit_03 · Unit 3 · Problems on finding the solution when RHS=0

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

Contents

1. 1 · Problems on finding the solution when RHS=0
2. Appendix A · Symbol glossary
3. Appendix B · Sources, coverage and gaps
4. Appendix C · Revision and retrieval practice

M1

Problems on finding the solution when RHS=0

1 Problems on finding the solution when RHS=0

Study focus: Problems on finding the solution when RHS=0. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000150 · EV0000151

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we'll solve problems on finding the general solution of homogeneous linear partial differential equations with constant coefficients.

Key details

- In the last class, we understood how to factorize.
- F of D comma D dash Z equals zero into linear factor that is AID plus AI D dash C i.
- Sometimes CI may be zero or B I may be zero or AI may be zero.
- if we get to know that the total degree of each term is the same, then we can use this shortcut to make the problem simple.
- And it clearly shows that the total degree of each term is M.
- Either we can use d by d dash to be m or we may look for.
- A plus b ol square or a minus b whole square that makes the problem even simpler.
- And this is same as 2ab that is d square and d dash square.
- A plus B and A minus B.
- The coefficient of d dash is minus 1, so b1 is minus 1, and there is no c.
- Yeah, so C1 is 0.
- As z equals x of phi 1 x plus y and phi 2 of x plus y.
- This is for the first factor.
- And this is for the second factor.
- We know a logic when we have both the terms to be negative, we take we can take minus common and that is same as this.
- Before that we introduced our.
- And now we get the three of D by D dash plus one of Z.
- Minus 3m plus 1 0.

- So we need to find out what are the three roots of m because this is a cubic equation in M.

Example or application from the lecture

- Let's look at the first problem d power four minus two d square t dash square plus d dash power four of the d plus zero.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000150 · EV0000151 · EV0000152 · +1 source units in pack.json

Core explanation · 05:00–09:59

So when we use synthetic division we see that m equals minus 1 is the first root by inspection and then we get to know that we get 4m square minus 4m plus 10.

Key details

- And this is same as 2m minus 1 whole square zero and we see m equal to 1 by 2 two times.
- and when we put m equal to half we get d by d dash equals one by two this is same as 2d equal to d dash so this is same as 2d minus d dash and this square because m equals half repeated two times from here.
- But the first.
- D plus d dash A B C R 110 respectively and for the second ABC R 2 minus 1 and 0 respectively.
- Whenever we see B is negative, we need to go for star concept, what we know from earlier.
- So for the first factor this is the solution for the second factor this is the solution.
- And you can see the order of equation is three.
- And the number of orbital functions are also three, therefore this is a general solution written as Z equals this.
- So when we use the shortcut method we see that the right equation in m is m square minus m minus x zero so m values three and minus two so obviously when we say a by d dash equals three the first factor will be d minus three d dash.
- And the second factor is d plus 2d dash.
- And these are the abc value C0.
- Then we get the solution to be phi one of 3x plus y, phi 2 of 32x minus y.
- In a similar manner if we look at the port problem, it is easy to see that and this part is nothing but a square sorry a minus b whole square.
- That is a d minus d dash square and we have another square outside.

Example or application from the lecture

- Let us substitute m as 1 minus 1 and 1 by 2 respectively in d by d dash.
- So let's first find out a i, bi and c i values.
- Let's look at the third problem that is d squared minus d dash minus six d dash square.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000154 · EV0000155 · EV0000156 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Problems on finding the solution when RHS=0 transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Problems_on_finding_the_solution_when_RHS_0__0bbcd9c646bfb2ec.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on finding the solution when RHS=0 transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Problems_on_finding_the_solution_when_RHS_0__0bbcd9c646bfb2ec.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on finding the solution when RHS=0 transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Problems_on_finding_the_solution_when_RHS_0__0bbcd9c646bfb2ec.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9965; only the JSON segmentation supplies the displayed timestamps.
Problems on finding the solution when RHS=0 transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Problems_on_finding_the_solution_when_RHS_0__0bbcd9c646bfb2ec.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
2. 1 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
3. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
4. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
5. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
6. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
7. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00046	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0037. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00047	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0037. Sidecar variants compare...
SRC00048	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0037. Sidecar variants compare...
SRC00049	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0037. Sidecar variants compare...

C Revision and retrieval practice